

# Exploring the effect of simulated crowd noise on multiple object tracking performance in usport football athletes

SPORT PSYCHOLOGY ABSTRACTS

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Rob McCaffrey\*, Kim Dorsch\*

## Abstract

Background noise is present in most daily activities. For some individuals, this noise can be dismissed; for others however, background noise can have a negative effect on an individual's ability to perform (e.g., SÅ¶derlund et al., 2010). Little work has been conducted on how attentionally-based performance (i.e., multiple object tracking, MOT) is affected when audio stimuli is present in athletic populations. The objective of this study was to examine if and how noise impacted MOT performance in university level football players. Twenty USPORT level football athletes (M age = 20.45 yrs, SD = 1.65 yrs) participated in a 6-session (18 trials) 3D-MOT training experience using the NeurotrackerTM (Cognisens Inc.). Ten athletes completed the training in a dark room with no external noise (had noise cancelling headphones); while the other ten athletes completed the training in the same room but were exposed to a consistent simulated crowd noise. No significant differences in baseline visual tracking speed (VTS) scores between the two groups ( $p > 0.05$ ) were found. After the 18 training sessions, the mean VTS score for the noise group was 2.07, SD = 0.24. The no noise group averaged significantly slower,  $t(1, 18) = 2.4$ ,  $p < 0.028$  at  $M = 1.77$ , SD = 0.32. Athletes typically perform in loud stadiums and these findings could be explained by the ability to block out external distractions. Indeed, the presence of the simulated crowd noise may actually enhance the ecological validity of the training sessions. Limitations and future directions will be discussed.

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